10K

f_1	f_2	f_3	f_4	Y
–	–	–	–	Yes
–	–	–	–	NO
–	–	–	–	Yes
–	–	–	–	NO
		⋮		⋮

$\xrightarrow{DT}$

f_1

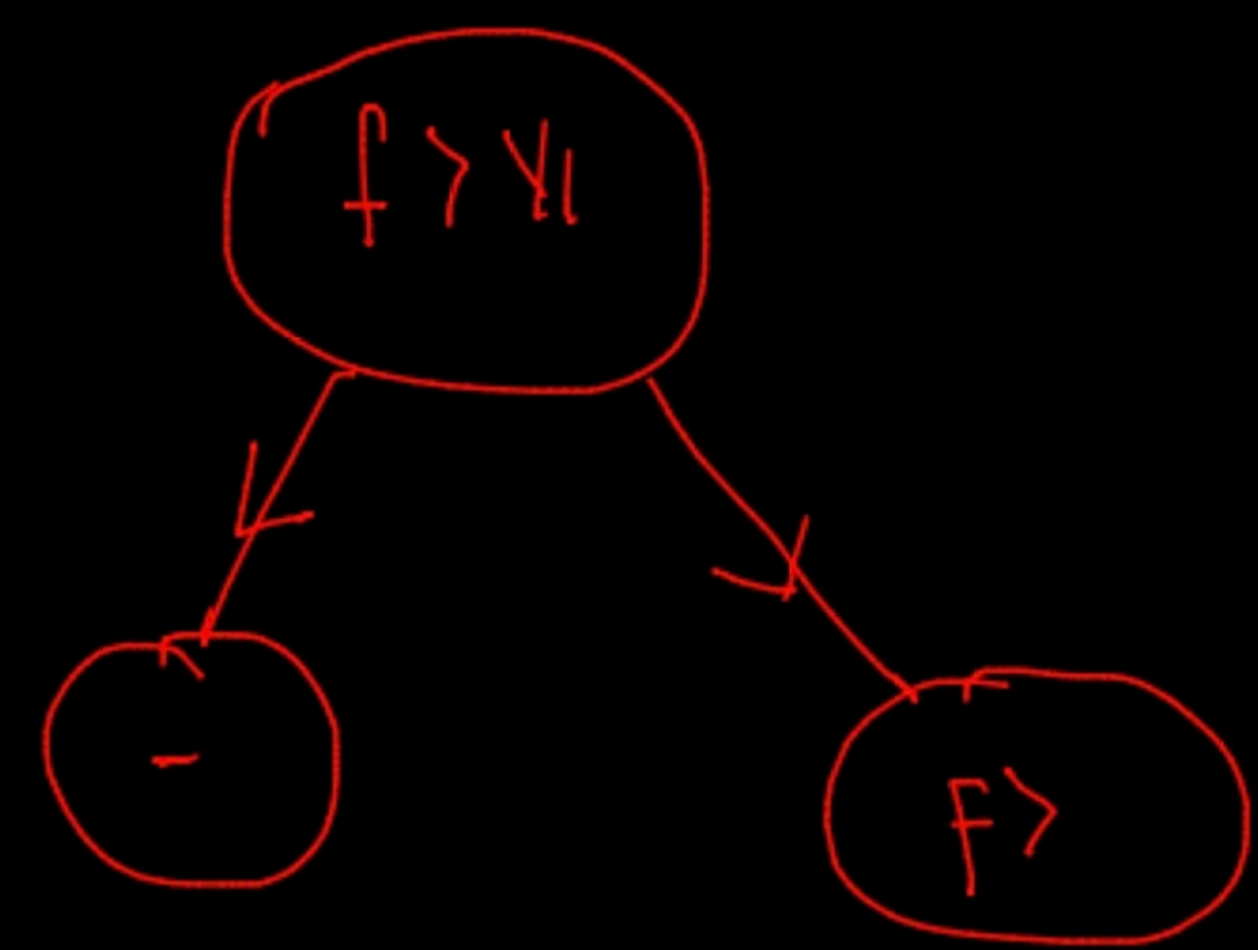
f_2

$f_3 \rightarrow$ Decide ?

f_4

measure the randomness $\leftarrow$ ENTROPY

IG



1 / 1

$H(Y) \Rightarrow 0.94$ → Dataset (14 points)

Outlook $\xrightarrow{(IG)}$ {outlook, playtennis}

D1, D2, D3

$H(D1) = 0.97$

$H(D2) = 0$

$H(D3) = 0.97$

⇒ weighted entropy

$$\frac{5}{14} * 0.97 + 0 + \frac{5}{14} * 0.97$$

weighted ENTROPY

$$\Rightarrow 2 * \frac{5}{14} * 0.97$$

Blackboard

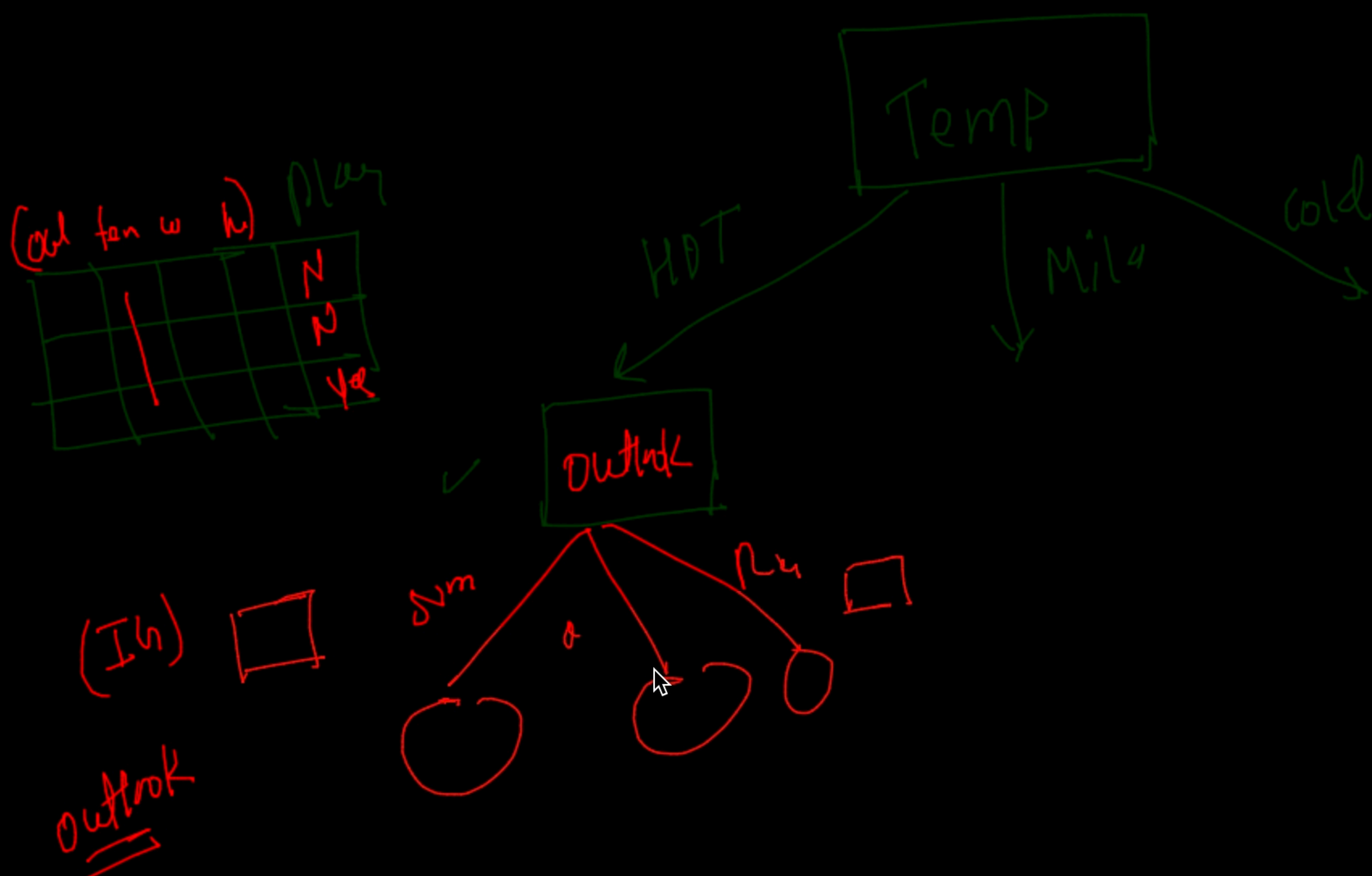
2 / 3

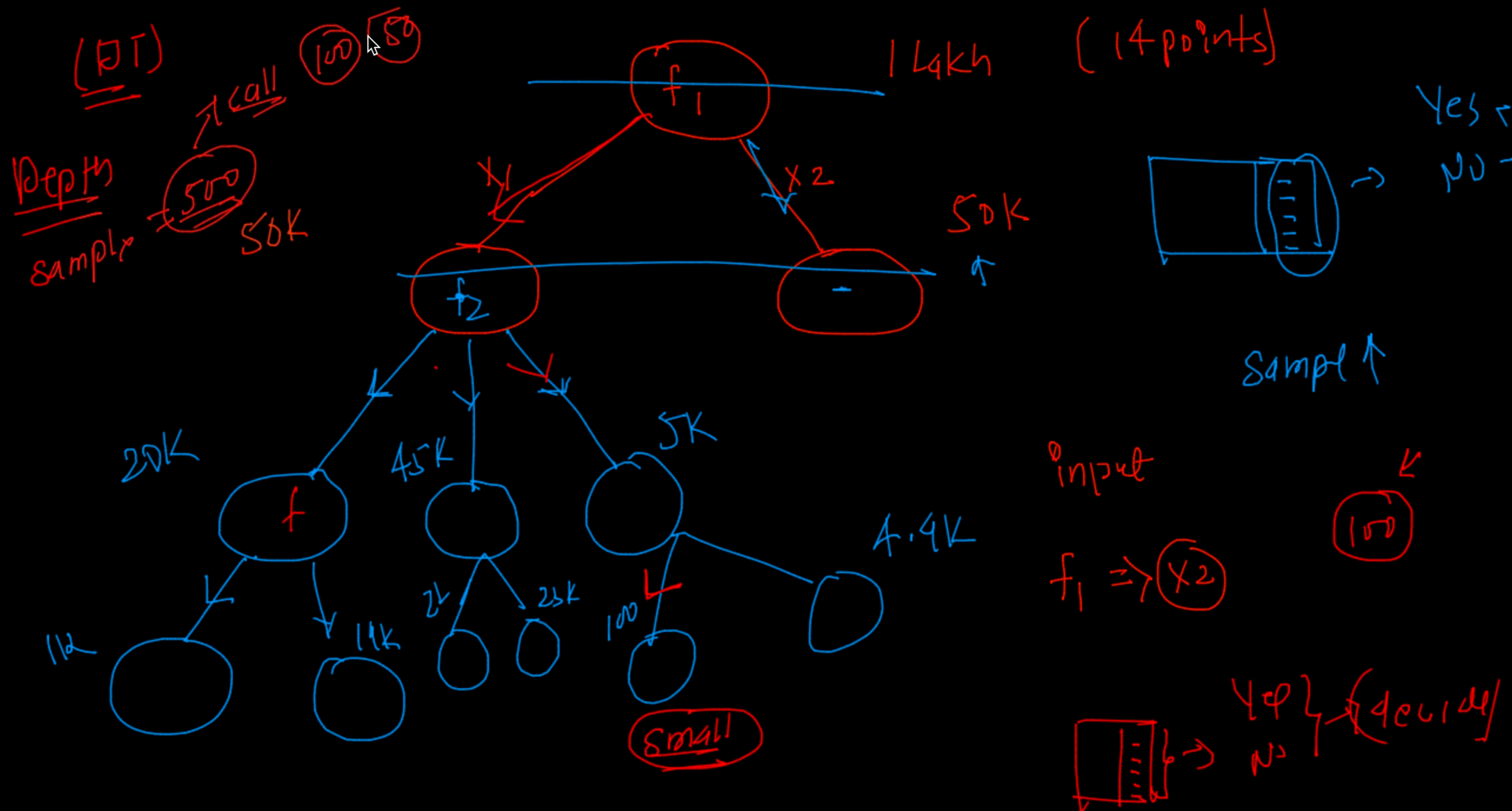
$$IG(Y, \text{Dataset}) = \left(\begin{array}{l}\text{weighted} \\ \text{entropy} \\ \text{after breaking} \\ \text{dataset into D1} \\ \text{D2 D3}\end{array}\right) - \left(\begin{array}{l}\text{Entropy} \\ \text{before} \\ \text{breaking} \\ \text{the data}\end{array}\right)$$

$$= 0.69 - 0.94$$

$$\boxed{IG(Y, \text{ud}) = \quad}$$

$$\Rightarrow \left.\begin{array}{l} IG(\text{outlook}) = 0.23 \\ IG(\text{temp}) = 0.24 \end{array}\right\} \quad \begin{array}{l} IG(\text{wind}) = 0.11 \\ IG(\text{hu}) = 0.21 \end{array}$$


Temp
HOT
Mild
Cold
Outlook
Play
N
N
Yes
Outlook


(DT)
100
50
recall
Depth
500
sample
50K
f1
1 Lakh
(14 points)
Yes
No
x1
x2
50K
f2
20K
45K
5K
f
4.4K
small
input
f1 => x2
100
Sample
decide

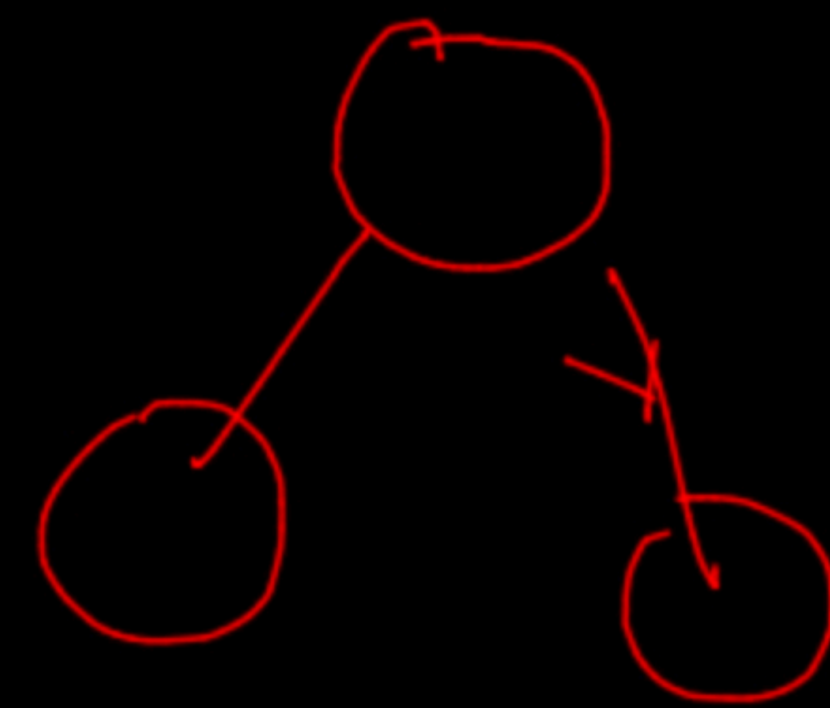
overfitting and underfitt

1 lakh

Decision stump

$h=1$

50K



50K

→ underfitting

→ yes / no

Hyper-parameters

⇓

h

over

less → → sample

IG

→ Gini impurity

(formula)

(case ii)

7

{4 yes, 3 No}

Blackboard